Unit V: Plant Physiology (Functional Organisation)

Chapter 12

Mineral Nutrition

Learning Objectives

The learner will be able to,

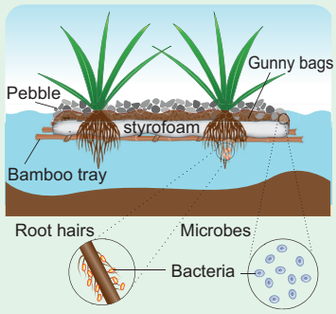
- *Recognise the need of mineral nutrition.*
- *Analyse the classification and criteria for essential minerals.*
- *Learn the techniques of Hydroponics and Aeroponics.*
- *Correlate different types of special modes of nutrition.*
- *Ability to recall and analyse nitrogen fixation.*

Chapter Outline

12.1 Classification of Minerals
12.2 Functions, mode of absorption, deficiency symptoms of Macronutrients
12.3 Functions, mode of absorption and deficiency symptoms of Micronutrients
12.4 Deficiency Diseases and symptoms
12.5 Critical Concentration and Toxicity of minerals
12.6 Hydroponics and Aeroponics
12.7 Nitrogen Fixation
12.8 Nitrogen Cycle and Nitrogen Metabolism
12.9 Special Modes of Nutrition

64OC1

A solution to Pollution



A new solution has come up for high nutrient pollution and eutrophication in surface waters. Floating Treatment Wetlands (FTWs) offer promising solution and it is a built structure which measures around 3,000 sq.ft and comprises four layers: floatable bamboo at base, styrofoam second layer, a third layer of gunny bags with gravels and final layer to support cleaning agents (plants). Native plants including Vetivers, *Citronella*, Tulsi and *Withania* are being researched for use as cleaning agents. FTW works on the principle of Hydroponics which is explained in this chapter. Microbes grown on the roots of these plants break down and consume organic matter in water and reduce pollution.

As a traveller you would have got a chance to observe the plants. It is an interesting fact that all plants are not alike. Just spend some time to observe the nature. You can notice plants with attractive leaves, flowers and fruits.

Can you say all plants are healthy and uniform in growth? Some plants are not healthy and show symptoms like texture changes, stunted growth, chlorosis, necrosis and so on. Can you tell what is the reason

XIth_Botany_Chapter_12_EM.indd 259 05-01-2023 18:29:32